

Technology Stack

Date	04 July2026
TeamID	
ProjectName	A Comprehensive Measure of Well-Being
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Fill out the table below to detail the specific technologies chosen for each component of your architecture and provide a brief justification for your choice.

Technology Stack Details

S.No	Architecture Component/Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	HTML, CSS, Bootstrap, JavaScript	Used to create a responsive and user-friendly interface where users can enter HDI indicators and view prediction results.

2	Backend / Server-Side	Python, Flask	Flask provides a lightweight web framework to process user inputs, communicate with the machine learning model, and return HDI predictions efficiently.
3	Database/Data Storage	CSV Dataset (HDI Dataset)	Stores historical Human Development Index data used for training and testing the machine learning model. It is simple, lightweight, and suitable for this prediction project.
4	Cloud/Hosting/ Deployment	Render / PythonAnywhere / Localhost	Used to deploy the web application, making it accessible online with minimal configuration and maintenance.
5	VersionControl& CI/CD	Git & GitHub	Git is used for version control, while GitHub enables code collaboration, project management, and backup of the source code.

6	Third-Party APIs/ Other Tools	Scikit-learn, Pandas, NumPy, Joblib, Matplotlib	Scikit-learn is used to train the machine learning model, Pandas and NumPy handle data processing, Joblib saves the trained model, and Matplotlib is used for data visualization and analysis.
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